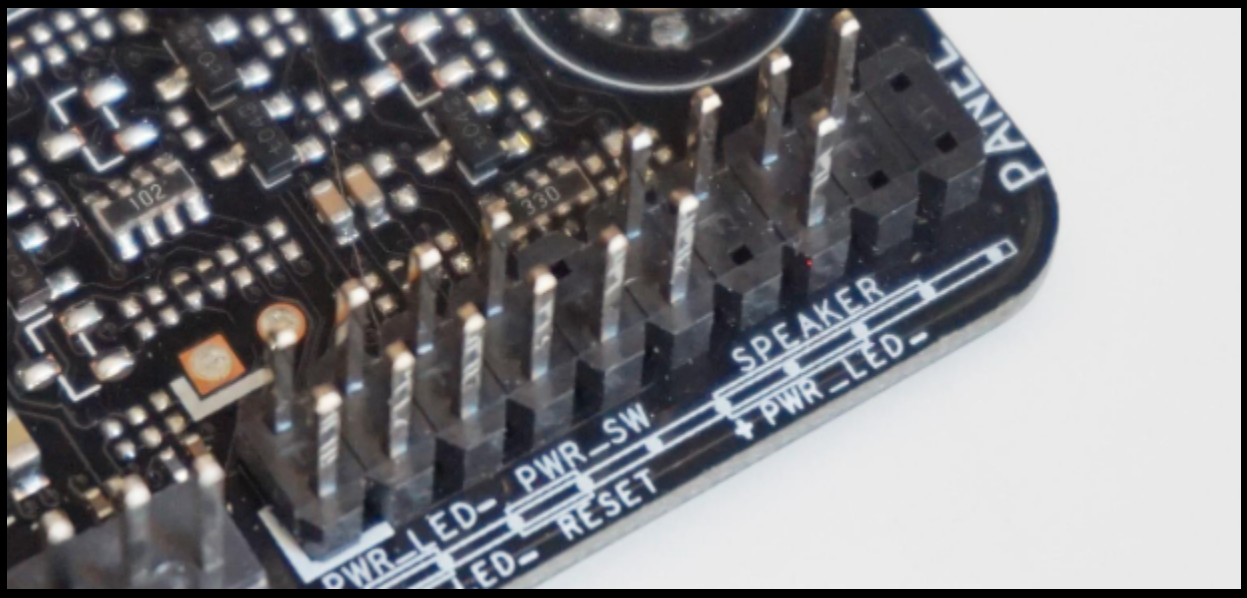
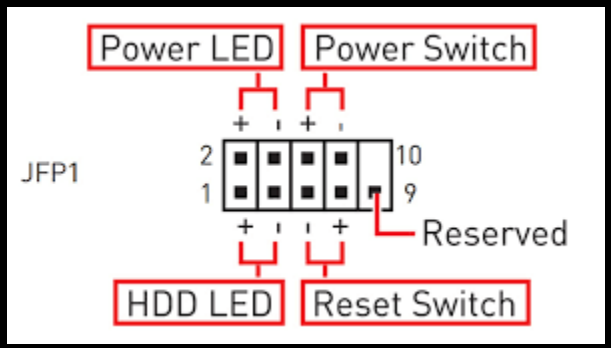
Computer Error Identification

Taking a computer apart is the easy thing to do. Putting it all back together successfully is way more of a challenge. Some things to consider when assembling and reassembling a computer:

- Make sure the power cable is disconnected
- Wear a static discharge wrist strap or discharge yourself by touching a non painted metallic part of the case
- Gather the correct tools (usually the phillips screwdriver is all you'll need but a flat head and torx screwdriver may be needed depending on the computer)
- Never force anything. If it's not coming out easily then you are probably doing something wrong or there are steps/tabs/screws to be pushed prior to removal.
- Take pictures to help you remember where things go
- 99% of all cable connectors are keyed so that they can only go in a certain way back into their header/socket (see image at the end of this document)
- Some are not keyed ex. Front panel connector. In this case you will need to look for markings/text on the connector and match them with those on the motherboard (take a picture too)







- Make sure connectors are are in their header/socket all the way
- Some components are very sensitive to damage, particularly the CPU. Make sure to drop this in its socket and NEVER push on it.

- Don’t forget to apply thermal paste to the CPU surface prior to mounting the heatsink on top of it

Despite these considerations people will often still make mistakes. Trying to turn a rebuilt computer back on and not having it restart is worrisome and at times costly. In this exercise we will try to identify the signs our computers give us when telling us there is a problem. Normally there are three methods for conveying a problem: 1. Beeps 2. On screen message and 3. Flashing LEDs on the front panel.

Please note that parts do not have to be mounted to test things. Just make sure nothing is hanging to the point it will damage the part or the cables connected to it.

Exercises:

9Remove the side or top case carefully.
Discharge your static electricity properly.
Connect your computer to power. Move the monitor off the top of the computer and put it aside (everything should still be connected).
The monitor should be on.
Turn your computer on. What do you notice happens to the insides of your computer when the power is turned on?
Log off windows and power down your computer. Disconnect the power connector from the back of the computer.
Find the HDD. It should have two cables connected. One for power and one for data. Identify each. Describe how their appearance differs.
Disconnect either cable. Reconnect the power and turn on the computer. Explain how your computer tells you that there is an issue regarding what you did.

In Bios it says “Alert Hard Drive not found”

Power off your computer. Disconnect the power cable. Plug back the HDD cable. Turn the computer on again and make sure everything works. When you know everything works, power it down again. This time find your computer’s memory (RAM). Remove all memory modules. Turn on your computer and explain what happens.

The monitor didn’t boot up and the computer starting beeping when we turned it on

Power off your computer. Disconnect the power cable. Replace the memory and verify the computer's operation. Note the key on the RAM stick and make sure you orient it correctly before inserting it back into the motherboard RAM sockets. Explain the key.

The RAM socket has a male female system when it’s fully centered so the RAM only goes in one way, the RAM being the female and the RAM slot being the male.

Power off your computer. Disconnect the power cable. Find the power cable connected to the CPU fan and disconnect it from the motherboard. Turn on the computer and describe what happens.

It gives a message saying“Alert CPU 0 fan failure”

Reconnect the CPU fan power cable connector. Explain the key. Reconnect the power and start the computer to verify it works.

The CPU fan able has a clip on one side that only goes in 1 direction with the female Connector on the motherboard.

Power off your computer. Disconnect the power cable. Find the large motherboard ATX power connector. Disconnect it. Reconnect the power cable. Try starting the computer. Explain what happens.

When we disconnected it, the computer doesn't power on, absolutely nothing happens, This is because the ATX power connector gives power to the motherboard, which guides power to every component in the computer.

Power off your computer. Disconnect the power cable. Reconnect the ATX power connector. Make sure your computer works. Find the switch/power cables that connect your front panel connector to the motherboard. Disconnect and test what happens. Explain it here.

Alert! Found I/O Cable Failure

Finally, find and remove the CMOS battery. Test and explain what happens.

The computer still worked and turned on but loses the stored system time ,date and bios Settings,as it reverts to default

Find the front panel switch connector and detach it from the motherboard. Plug the power cable back and try starting the computer. What happens?

Alert! Found I/O Cable Failure

Remove the CPU from the MB. Place it carefully on the table (not on a metallic surface). Put back the CPU fan and heatsink. Try to start the computer. Explain what happens.

CPU fan runs,the power button is orange no boot

18. Try removing one of the ODD SATA cables and explain what happens when you start the computer.

Our computer doesn't have ODD

19. Remove an ATX power connector and test the response to restarting the computer.

When we disconnected it, the computer doesn't power on, absolutely nothing happens, this is because the ATX power connector gives power to the motherboard, which guides power to every component in the computer.

20. Fill in the following table so that you describe how the computer warns you of the issues described in the second column.

Error Identifier	Error Description
Explain what the computer does to alert you to this problem	**The following is a brief description of the problem.**
1. In the bios it says “Alert, Hard Drive not found!”	HDD cable is disconnected or there is a general failure of the HDD.
2.Beeping comes from computer with no display or boot	RAM is missing, not seated properly or has failed.
3.In Bios it says “Alert CPU 0 fan failure”	CPU fan power cable is not connected properly. CPU is overheating.
4. No boot or any sort of response of anything at al	ATX power cable is not connected. MB is not getting power.
5.	There is an issue with the main power switch.
6.Computer will still turn on but loses stored system of time and date in bios setting as it returns to default	There is an issue with the system or CMOS battery.
7.CPU fan runs, power button is orange, no boot	There is an issue with the CPU.
8.We don't have a ODD	ODD SATA cable removed/loosened.
9.When we disconnected it, the computer doesn't power on, absolutely nothing happens, this is because the ATX power connector gives power to the motherboard, which guides power to every component in the computer.	ATX power connector removed/loosened.

The last thing you need to do is to find the online manual/documentation that describes all the possible error codes your computer can generate. Find out your computer model (see the front panel for this info). Perform a web search and list the URL to this information below:
Computer Model:

Dell Optiplex 790

Error Codes URL:

https://dl.dell.com/manuals/all-products/esuprt_desktop/esuprt_optiplex_desktop/optiplex-790_owner%27s%20manual2_en-us.pdf
Page 71
Search through the documentation and see how your results aligned with what is described for the errors you tested.

Ultimate Chart of Computer Connectors and Ports

Updated for 2020

USB, Keyboard and Mouse


USB 2.0 Type A (male)


USB 2.0 Type A (female)


USB 2.0 Type B (male)


USB 2.0 Type B (female)


USB 2.0 Mini-A (male)


USB 2.0 Mini-B (male)


USB 2.0 Micro-A (male)


USB 2.0 Micro-B (male)


USB 3.0 Type A (male)


USB 3.0 Type A (female)


USB 3.0 Type B (male)


USB 3.0 Micro B (male)


USB 3.0 Type C (male)


USB 3.0 Type C (female)


PS/2 (male)


AT Keyboard (male)

Storage / Disk


SATA Type A


e-SATA


mSATA



M.2


U.2


SAS (Serial Attached SCSI)

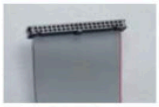

Lightning


Thunderbolt


Firewire 400 1394a 6-pin


Firewire 400 1394a 4-pin


Firewire 800/3200 1394b/c


IDE 40-pin


SCSI Micro DB68 (female)


SCSI Micro DB50 (male)


Internal 50-pin SCSI (f)


Internal 68-pin SCSI (f)

Network / Communications


GigE RJ-45 (SFP)


Ethernet RJ-45


Modem/Phone RJ-11


Serial/Parallel (f) DB-25

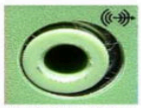

Serial RS232 (m) DE-9


Serial RS232 (f) DE-9


Centronics Parallel 36-pin


MIDI/gameport/DA-15

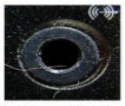
Audio


Line-Out, Headphones

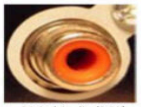

Microphone


Stereo Line In


Subwoofer/Center Out


Rear Surround Speakers


Mid Surround Speakers


Digital Audio (RCA)


Optical Audio "Toslink"

Video


VGA/SVGA (male) DE-15


VGA/SVGA (female) DE-15


DisplayPort (male)


DisplayPort (female)


Mini DisplayPort (m)

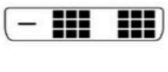

Mini DisplayPort (f)


HDMI Type A (male)


HDMI Type A (female)


DVI-I Single Link


DVI-I Dual Link


DVI-D Single Link


DVI-D Dual Link LFH-60


DVI-D Dual Link DMS-59


Micro-DVI


Mini-DVI


Mini HDMI Type C


Composite A/V (RCA)


S-Video


Component Video (m)


Component Video (f)


RF/Coaxial (male)


RF/Coaxial (female)


Apple Display Conn (ADC)


Micro HDMI Type D

Power


IEC 320 C13/C14 Connector


IEC 320 C13 Socket (f)


IEC 320 C14 Plug (m)


IEC 320 C19 (f)


IEC 320 C20 (m)


IEC 320 C5 Connector


SATA Power Connector


Molex 4-pin Connector

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